

# SIMULTANEOUS IDENTITIES OF THE GOLDEN RATIO AND EULER’S IDENTITY, FORMALLY VERIFIED IN LEAN 4 OVER THE KNOWN MERSENNE PRIME EXPONENTS

**ABSTRACT.** We present a sustained study of a single object followed across representations, where the content lives in the equivalences between them, not in any representation in isolation. We prove and formally verify in Lean 4 that the golden ratio  $\varphi = (1 + \sqrt{5})/2$  satisfies simultaneously twelve identities, inclusions, and identifications across five canonical structures: the complex rotor of Euler’s identity, the real trigonometric and real-multiplicative lines, the arithmetic of Mersenne numbers modulo 20, and the Galois group  $(\mathbb{Z}/20\mathbb{Z})^\times$  of  $\mathbb{Q}(\zeta_{20})/\mathbb{Q}$ . The central clause is the identity  $3\varphi^{\sigma(p)} = 2^p$ , with  $\sigma(p) = p\lambda - \log_\varphi 3$  and  $\lambda = \log_\varphi 2$  transcendental by Gelfond–Schneider. Specialised to the 52 known Mersenne prime exponents (GIMPS, 1952–2024), it holds exactly for each, and every pair of Mersenne numbers, in their binary form  $2^p = M_p + 1$ , stands in an exact golden ratio. The factor 3 coincides with  $M_2$ , with the number of admissible residue classes of  $M_p \bmod 20$ , and with the numerator of  $M_2/|(\mathbb{Z}/20\mathbb{Z})^\times| = 3/8$ . The verification is conducted against Mathlib4 with 0 `sorry`, taking as external axioms only the Gelfond–Schneider theorem and the 44 large GIMPS primalities.

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## 1. INTRODUCTION

**1.1. Classification under equivalence: a millennial question.** The question of which discrete objects fit canonically inside which continuum, and how the continuum organises them by its symmetry, is older than its modern formalisation. Vitruvius (*De Architectura*, c. 30 BC) inscribes the human body simultaneously within a square and a circle sharing a common centre: the body itself plays the role of *modulus*, a system of an origin, a radius, an angle, and a magnitude, mediating between the square as discrete arithmetic substrate (four elements, four cardinal directions, the binary doubling  $1 \mapsto 2 \mapsto 4$ ) and the circle as continuous rotational symmetry. Cesare Cesariano's 1521 commentary on Vitruvius records the same module geometrically through a sequence of nested regular polygons sharing axis (the origin) and base vertex (the radius at angle zero), from triangle to dodecagon: each polygon discretises the circle at the angle  $2\pi/n$ , all fitting inside the same continuous rotational ambient. The polygonal sequence is a geometric record of a structural question: given an origin, a radius, and a magnitude, which angular discretisations admit canonical inscription inside the circle's rotational symmetry, and in what form.

Arguably the evolution of the same ancient structural question is taken up in two foundational modern treatises. Euler's *Introductio in analysin infinitorum* (1748)—characterised by Boyer at the 1950 ICM as “the foremost textbook of modern times” [1]—establishes the function as the central object of analysis and gives a systematic analytic treatment of the exponential and trigonometric identities, including  $e^{i\theta} = \cos \theta + i \sin \theta$ —an analytic rotor of origin 0, radius 1, angle  $\theta$ , and unit magnitude—whose inverse form  $i\theta = \log(\cos \theta + i \sin \theta)$  had been known to Cotes in 1714. Gauss's *Disquisitiones Arithmeticae* (1801, drafted from 1796) gives the systematic arithmetic counterpart, consolidating the work of Fermat, Euler, Lagrange, and Legendre into a unified framework—including in particular the arithmetic of the cyclotomic roots  $\zeta_n = e^{2\pi i/n}$ , the angles  $2\pi/n$  of the regular  $n$ -gon treated as algebraic integers—that, in Kline's phrasing, “began the modern theory of numbers” [2] and set the directions of subsequent work in the subject.

**1.2. Euler 1748: the analytic-geometric substrate.** The identity

$$e^{i\theta} = \cos \theta + i \sin \theta \tag{1.1}$$

defines the one-dimensional complex rotor: cosine and sine are the Cartesian projections of  $e^{i\theta}$  onto the real and imaginary axes. The unit circle  $S^1 \subset \mathbb{C}$  is the locus of  $e^{i\theta}$ , and  $2\pi$  is the length of one revolution. Within  $S^1$ , the  $n$ -th roots of unity  $\zeta_n = e^{2\pi i/n}$  are the vertices of the regular  $n$ -gon: the polygonal discretisation of the circle *is* the theory of roots of unity. Cesariano's nested polygons are the geometric trace of what (1.1) encodes analytically.

The identity  $e^{i\pi} + 1 = 0$  is the evaluation of (1.1) at  $\theta = \pi$ : an exact identity in  $\mathbb{C}$  in which several independently defined objects—the analytic exponential, the algebraic unit  $i$ , the geometric constant  $\pi$ , and the arithmetic units 1 and 0—coincide simultaneously, in a way not implied by their definitions taken individually. The content of (1.1) is this simultaneous coincidence of independent identities at a single object.

Subsequent mathematical work over the analytic-geometric substrate of (1.1) has produced further simultaneous identities and related results of varying form (reciprocity laws, congruences, equivalences, motivic realisations). Immediately after Euler, Gauss (*Disquisitiones Arithmeticae*, 1801) established structural identities connecting the roots of unity defined by (1.1) to arithmetic

data modulo primes, opening the systematic arithmetic exploitation of the substrate. Over the two centuries that separate Gauss from the present, this line has multiplied through successive nineteenth- and twentieth-century developments; an exhaustive account is beyond the scope of this note. Several examples illustrate the range. In quantum mechanics, the Feynman path integral expresses the propagation amplitude as a sum  $\int e^{iS[\gamma]/\hbar} \mathcal{D}\gamma$  in which the Eulerian exponential of the action governs the analytic structure of the theory. In harmonic analysis on locally compact abelian groups (Pontryagin), the characters  $\chi_\xi(x) = e^{2\pi i \langle x, \xi \rangle}$  realise the duality between a group and its dual; the parallel theory of unitary representations of compact Lie groups proceeds through characters valued in the unit circle. The Hilbert–Pólya conjecture proposes that the nontrivial zeros of  $\zeta(s)$  are the eigenvalues of a self-adjoint operator  $H$ , whose associated unitary group  $\{e^{itH}\}_{t \in \mathbb{R}}$  would realise the spectrum as a one-parameter rotation; the Selberg trace formula relates the spectrum of the Laplacian on a hyperbolic surface to its closed geodesics through analogous exponential sums. In arithmetic geometry, the Weil cohomologies place the roots of unity  $\zeta_n = e^{2\pi i/n}$  at the centre of the Frobenius action on étale cohomology, and Grothendieck's theory of motives organises periods  $\int_\gamma \omega$ —including the elementary  $\oint dz/z = 2\pi i$ —as fundamental invariants. More recently, identifications of unification across arithmetic, tropical geometry, and motivic theory have been pursued over the  $\mathbb{F}_1$  substrate (Connes–Consani, 2014–2021 [3]).

Across these developments, arithmetic content otherwise undecidable in the sense of Gödel [4] becomes accessible through the analytic-geometric substrate of (1.1): Tarski [5] showed that elementary geometry over the real-closed field admits a decision procedure, by contrast with elementary arithmetic. The field-with-one-element programme initiated by Tits [6] and developed by Manin [7] among others pursues a geometry over a hypothetical “field with one element”, under which  $\mathbb{Z}$  would behave as a curve and roots of unity supply the arithmetic with  $\mathbb{F}_1$ -structure, so as to carry methods native to geometry over finite fields into the study of  $\mathbb{Z}$ .

This aim descends from Weil's “Rosetta Stone”. In a 1940 letter to his sister Simone [8], Weil set out an analogy among three domains—number fields, function fields over a finite field, and Riemann surfaces—read as three columns of a single text. In the middle column he was able to prove the Riemann hypothesis in its form for curves over finite fields [9]; the ambition of the field-with-one-element programme is to endow  $\mathbb{Z}$  with enough geometry that this proof, or an analogue of it, may be transported to the number-field column and settle the classical Riemann hypothesis. The Rosetta Stone is itself an instance of the classification under equivalence of § 1.1: one arithmetic content read across representations that are not identical but structurally the same, the decisive work lying in the translation—the equivalences—between them.

The present note operates from a particular angle on this substrate. In some sense closer to the modulus of § 1.1 than to those modern extensions, it uses the golden ratio  $\varphi = (1 + \sqrt{5})/2$  together with the rotor of Euler's identity to articulate identities between the analytic and the arithmetic through the geometric structure of  $\varphi$ , exploring the relations that  $\varphi$  admits between  $e^{i\theta}$ ,  $\mathbb{R}$ , and  $\mathbb{C}$ .

In what follows the golden ratio  $\varphi$  enters the argument as a structural operator between discrete polygonal geometry—specifically the pentagon and decagon—and the continuous rotational symmetry of the unit circle. The relations exploited here are inspired by the proportional compass of classical artistic practice, their use of ratios based on the human body and the much later invention of the *sectio aurea* tool: not as an artisan's empirical approximation, but as an exact coupling between geometric magnitudes and algebraic roots. Grisales Herrera [10] establishes one face of this coupling between  $\varphi$  and the integers via the eight reduced residue classes of  $(\mathbb{Z}/20\mathbb{Z})^\times$ ; a parallel angular-rotation framework for primes is developed by Acevedo Agudelo [11]. The present note records, in the analytic formulation of Euler's identity (1.1), the broader configuration of which these facts are part.

For the golden ratio, the starting point is geometric: among the vertices  $\{\zeta_{10}^k\}_{k=0}^9$  of the regular decagon defined by (1.1) at  $n = 10$ , the sum of two conjugate vertices equals  $\varphi$ ,

$$\varphi = \zeta_{10} + \zeta_{10}^{-1}, \quad \zeta_{10} = e^{i\pi/5} \quad (\text{clause (i) of Theorem 1.1}).$$

This inscription places  $\varphi$  inside the structure of (1.1) as a canonical point. Three further inscriptions follow from this one and constitute the content of the present note. First, the real part of the same vertex sum gives the trigonometric identity  $\varphi = 2 \cos(\pi/5)$  in  $\mathbb{R}$  (clause (iii)). Second, the multiplicative line  $\mathbb{R}_{>0}$  admits a transcendental period  $\lambda_{\log} = \log_{\varphi} 2$  that plays the structural role of the  $2\pi$  of (1.1) and closes a turn between the algebraic objects  $\varphi$  and 2,

$$\varphi^{\lambda_{\log}} = 2 \quad (\text{clause (iv)}),$$

a real-multiplicative companion to (1.1). Third, the cyclotomic field  $\mathbb{Q}(\zeta_{20})$  where  $\varphi$  resides has Galois group  $(\mathbb{Z}/20\mathbb{Z})^\times$  of order 8 (clause (ix)), and the Mersenne primes  $M_p$  distribute over the structural fraction

$$M_2/|(\mathbb{Z}/20\mathbb{Z})^\times| = 3/8 \quad (\text{clause (xi)})$$

of those eight Galois classes. These five inscriptions, together with seven further identities, inclusions, and identifications in the same canonical structures (clauses (ii), (v), (vi), (vii), (viii), (x), (xii) of Theorem 1.1), are collected in the single formal statement Theorem 1.1 below and verified in Lean 4. The relevance of  $\varphi$  at this point in the argument is that the inscription  $\varphi = \zeta_{10} + \zeta_{10}^{-1}$  in (1.1) propagates—trigonometrically into  $\mathbb{R}$ , multiplicatively into  $\mathbb{R}_{>0}$ , and arithmetically into the Galois group of the cyclotomic extension—producing twelve coincident identities under a single algebraic object.

**1.3. A simultaneous identities theorem for  $\varphi$ .** By a *simultaneous identities theorem* we mean a statement asserting that a mathematical object satisfies simultaneously a finite collection of identities provable in distinct canonical structures, with the simultaneous validity of the collection as the content of the statement. The corresponding statement for  $\varphi$  is the following Theorem 1.1. Its twelve clauses are proved in § 2 and § 3 (clauses (i)–(iii), (ix) in Remark 3.4, clause (iv) in Lemma 2.3, clause (v) in Theorem 2.2, clause (vi) in Theorem 3.1, clause (vii) in Theorem 3.3); the four remaining clauses (viii), (x), (xi), (xii) have short proofs but each registers a structural knotting articulated after the statement.

The individual proofs of the clauses are short—several require only definitional manipulation or elementary modular computation. The substance of the statement does not reside in the difficulty of the separate proofs but in the simultaneity of their validity under a single  $\varphi$ , and in the absence of a prior reference recording them jointly. The situation is analogous to that of Euler's identity  $e^{i\pi} + 1 = 0$ , whose modern proof is the evaluation of a series but whose enunciation revealed a knotting of analysis, algebra, geometry, and arithmetic not visible in the prior literature. The formal verification in Lean 4 secures the simultaneity as a single theorem, eliminating the possibility that the clauses hold separately but not jointly. Throughout,  $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$  denotes the canonical embedding and  $\varphi_{\mathbb{C}} := \iota(\varphi)$  its image in  $\mathbb{C}$  under this embedding.

**Theorem 1.1** (Simultaneous identities for  $\varphi$  (formal version: Theorem 3.5)). *A single  $\varphi$  satisfies simultaneously the following twelve identities, inclusions, and identifications, organised by the canonical structure they inhabit:*

In the complex rotor of Euler's identity  $\{e^{i\theta}\} \subset \mathbb{C}$ :

- (i) Cyclotomic inscription:  $\varphi_{\mathbb{C}} = \zeta_{10} + \zeta_{10}^{-1}$ , with  $\zeta_{10} := e^{i\pi/5}$ .
- (ii) Tenth-root closure:  $\zeta_{10}^{10} = 1$ .
- (xii) Rotor discretisation: for every  $k \in \mathbb{N}$ ,  $(\zeta_{10}^k)^{10} = 1$ .

In the real trigonometric line, with cross-identification to the complex inscription:

- (iii) Pentagonal trigonometric identity:  $\varphi = 2 \cos(\pi/5)$ .

(viii) Real-complex identification:  $\iota(\varphi) = \varphi_{\mathbb{C}}$ .

In the real-multiplicative line  $\mathbb{R}_{>0}$ :

(iv) Logarithmic binary bridge:  $\varphi^{\lambda_{\log}} = 2$ , where  $\lambda_{\log} := \log_{\varphi} 2$ .

(v) Universal real-multiplicative identity:  $\forall p \in \mathbb{R}, 3 \varphi^{\sigma(p)} = 2^p$ , where  $\sigma(p) := p \lambda_{\log} - \log_{\varphi} 3$ .

In the arithmetic of Mersenne modulo 20:

(vi) Modular concentration: *for every prime  $p$ ,  $M_p \bmod 20 \in \{3, 7, 11\}$ .*

(vii) Triple ternary identification:  $3 = M_2 = |\{3, 7, 11\}|$ .

In the Galois group of the cyclotomic extension  $\mathbb{Q}(\zeta_{20})/\mathbb{Q}$ , with quantitative ratio to the arithmetic locus:

(ix) Galois order:  $|(\mathbb{Z}/20\mathbb{Z})^{\times}| = 8$ .

(x) Galois sub-object inclusion:  $\{3, 7, 11\} \subseteq (\mathbb{Z}/20\mathbb{Z})^{\times}$ .

(xi) Quantitative ratio:  $|\{3, 7, 11\}|/|(\mathbb{Z}/20\mathbb{Z})^{\times}| = M_2/8 = 3/8$ , *equivalently*  $|\{3, 7, 11\}| \cdot 8 = M_2 \cdot |(\mathbb{Z}/20\mathbb{Z})^{\times}|$ .

No proper sub-collection of the twelve clauses constitutes the content; the content is the simultaneous holding of all twelve under a single  $\varphi$  and in relation to Euler's identity (1.1), where the constants  $\zeta_{10} = e^{i\pi/5}$ ,  $\lambda_{\log} = \log_{\varphi} 2$ , and  $\sigma(p) = p \lambda_{\log} - \log_{\varphi} 3$  are formally introduced in § 2 (Definition 2.1) and Remark 3.4. The five groupings above record the five canonical structures in which  $\varphi$  is independently inscribed: the complex rotor with its tenth-root discretisation, the real trigonometric line, the real-multiplicative line, the arithmetic of Mersenne modulo 20, and the Galois group of the cyclotomic decagonal extension. The cross-identifications (clauses (viii) and (xi)) record that the inscriptions are inscriptions of *the same*  $\varphi$  across distinct substrates and that the arithmetic locus sits as a quantitatively specific sub-object of the Galois group. § 3.1 makes this precise as a consistency theorem: the factor 3, the base  $\varphi$ , and the slope  $\lambda_{\log}$  are each pinned by a theorem external to the central identity, and the identity is the unique point at which they agree.

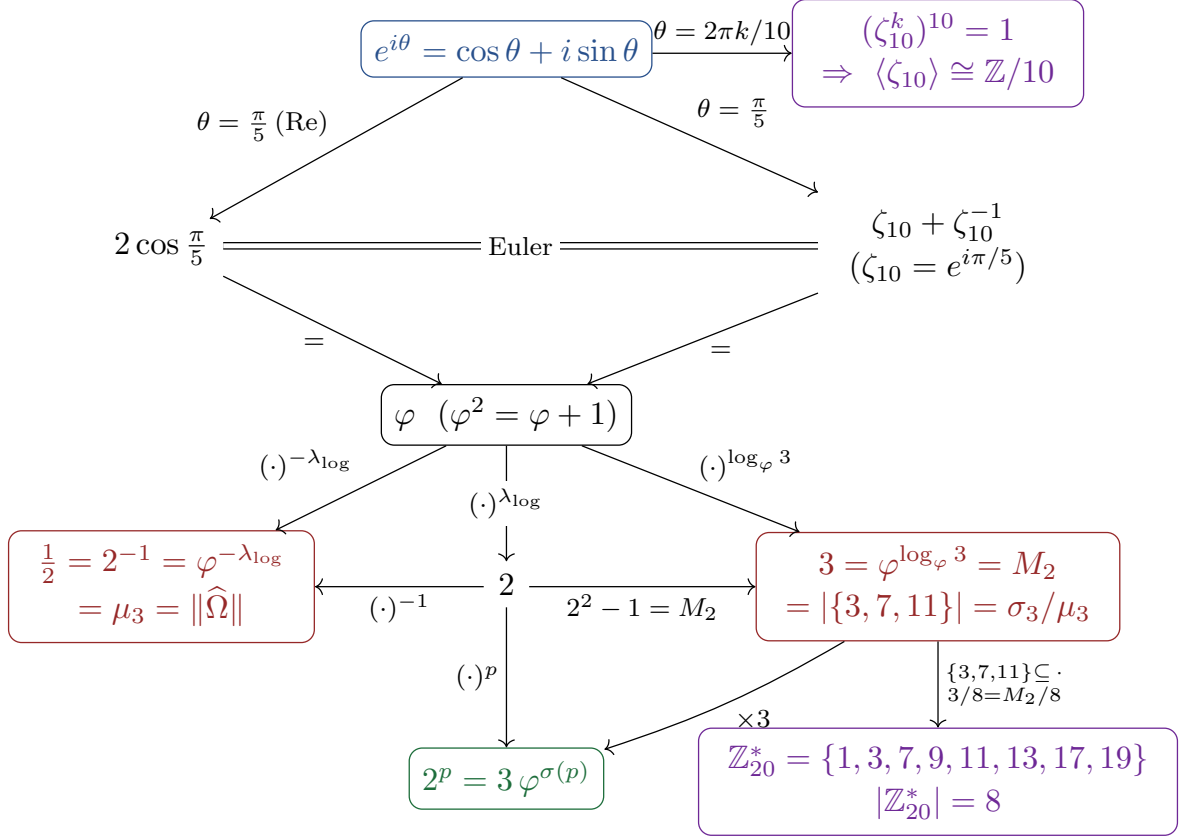


FIGURE 1. The twelve simultaneous identities of **Theorem 1.1** (clauses (i)–(xii)), together with the reciprocal turn  $\varphi^{-\lambda_{\log}} = \frac{1}{2}$  of § 5, as a single commutative diagram. Every directed path commutes to the universal identity  $3 \varphi^{\sigma(p)} = 2^p$  (green) and to the reciprocal turn  $\frac{1}{2} = \varphi^{-\lambda_{\log}}$  (red, left); the factor 3 =  $M_2$  and the half each appear as a convergence of several identifications under the single algebraic object  $\varphi$ .

Clauses (vi), (vii), (x), (xi), taken in isolation, fall in the genealogy of arithmetic identities over  $\mathbb{Q}(\sqrt{5})$  companion to Lucas's 1878 Fibonacci congruence and to Grisales Herrera's classification of the eight residue classes of  $(\mathbb{Z}/20\mathbb{Z})^\times$  into split and inert types in  $\mathbb{Q}(\sqrt{5})$  [10]. They enter **Theorem 1.1** as four of the twelve simultaneous clauses of the simultaneous identities theorem, not as the global content of the theorem. The splitting type of each prime  $M_p$  in  $\mathbb{Q}(\sqrt{5})$  is developed in § 4 via Lucas's congruence, where it is shown to be governed by the exponent  $p \bmod 4$  and to single out the inert primes as those satisfying the Lucas–Lehmer divisibility  $M_p \mid F_{M_p+1}$ .

Beyond these twelve simultaneous clauses, the same multiplicative period  $\lambda_{\log}$ , read in reverse, closes a reciprocal turn  $\varphi^{-\lambda_{\log}} = \frac{1}{2}$ . This value is not a thirteenth clause but a point already inscribed in the identity—the reciprocal of clause (iv), the  $p = -1$  node of the universal identity (v), and the multiplicative-line image of the rotor's conjugate root (clause (i))—developed in § 5, where it is also identified as the value at which the companion operator paper anchors its Hilbert–Pólya operator.

All statements of this note are formally verified in `golden_mersenne.lean`, a self-contained Lean 4 file built against Mathlib4 (toolchain v4.28.0) [12]; no `sorry` occurs in any proof. Two external inputs enter as axioms: the Gelfond–Schneider theorem (not yet available in Mathlib

mainline), which underlies the transcendence of  $\lambda_{\log}$ , and the 44 Lucas–Lehmer [13] primalities  $M_p$  prime for  $61 \leq p \leq 136\,279\,841$  (GIMPS [14]). The relation  $\varphi^2 = \varphi + 1$  is a theorem, not an axiom.

## 2. STATEMENT AND PROOF OF THE UNIVERSAL IDENTITY

Throughout this section and the next,  $\varphi^x$  for real  $x$  denotes the real exponential  $\exp(x \log \varphi)$ , the standard extension of integer exponentiation to  $\mathbb{R}$  via the natural logarithm of the positive real  $\varphi > 0$ .

The architectural pattern of (1.1)—a transcendental period mediating between algebraic objects—admits a second realisation on the multiplicative line  $\mathbb{R}_{>0}$ , where the binary base  $2^p$  takes the place of the rotor and the golden base  $\varphi^\sigma$  replaces the angular coordinate. The mediator is the constant

$$\lambda_{\log} := \log 2 / \log \varphi = \log_\varphi 2 \approx 1.4404,$$

transcendental over  $\mathbb{Q}$  by Gelfond–Schneider (Remark 2.9 below), playing the structural role of  $2\pi$ . It is the elementary period along which  $\varphi^x$  scales from 1 up to the next integer base 2:

$$\varphi^{\lambda_{\log}} = 2, \tag{2.1}$$

the *golden turn* (clause (iv) of Theorem 1.1). The multiplicative line  $\mathbb{R}_{>0}$  admits a second canonical scaling,  $\varphi^{\log_\varphi 3} = 3$ , and the composition of the two turns is the central identity of this section:

**Definition 2.1** (Golden index).  $\sigma: \mathbb{R} \rightarrow \mathbb{R}$ ,  $\sigma(p) := p \lambda_{\log} - \log_\varphi 3$ .

**Theorem 2.2** (Universal identity). *For every  $p \in \mathbb{R}$ ,  $3 \varphi^{\sigma(p)} = 2^p$ .*

This is clause (v) of Theorem 1.1. The multiplier 3 is not free: Theorem 3.3 below (clause (vii)) shows that it coincides with the first Mersenne prime  $M_2 = 2^2 - 1$  and with the cardinality of the admissible residue set of Theorem 3.1, and no other positive real admits this simultaneous identification. The proof of Theorem 2.2 is established below in three steps:

**Lemma 2.3** (Binary turn).  $\varphi^{\lambda_{\log}} = 2$ .

*Proof.*  $\varphi^{\lambda_{\log}} = \exp(\lambda_{\log} \log \varphi) = \exp(\log 2) = 2$ . Lean: `mersenne_bridge_via_lambda`. □

**Lemma 2.4** (Ternary turn).  $\varphi^{\log_\varphi 3} = 3$ .

*Proof.* Identical with 3 in place of 2. Lean: `phi_pow_log3`. □

*Proof of Theorem 2.2.* By definition  $\sigma(p) = p \lambda_{\log} - \log_\varphi 3$ , hence

$$\begin{aligned} 3 \varphi^{\sigma(p)} &= 3 \varphi^{p \lambda_{\log}} \varphi^{-\log_\varphi 3} && \text{(real-exponent subtraction)} \\ &= 3 (\varphi^{\lambda_{\log}})^p (\varphi^{\log_\varphi 3})^{-1} && (\varphi^{ab} = (\varphi^a)^b \text{ for } \varphi > 0) \\ &= 3 \cdot 2^p \cdot 3^{-1} && \text{(Lemmas 2.3 and 2.4)} \\ &= 2^p. && \square \end{aligned}$$

Lean: `golden_tower_bridge`.

**Theorem 2.5** (Uniqueness of the index). *For each  $p \in \mathbb{R}$ ,  $\sigma(p)$  is the only real  $\sigma$  such that  $3 \varphi^\sigma = 2^p$ .*

*Proof.* If  $3 \varphi^\sigma = 2^p$  and  $3 \varphi^{\sigma(p)} = 2^p$  then  $\varphi^{\sigma - \sigma(p)} = 1$ , hence  $(\sigma - \sigma(p)) \log \varphi = 0$ , and since  $\log \varphi > 0$  (as  $\varphi > 1$ ),  $\sigma = \sigma(p)$ . Lean: `sigma_mersenne_unique`. □

**Theorem 2.6** (Direct golden ratio between Mersenne). *For every pair  $(p, q) \in \mathbb{R}^2$ ,*

$$\frac{2^q}{2^p} = \varphi^{(q-p)\lambda_{\log}}.$$



Specialised to the GIMPS list, for every pair  $(p_i, p_j) \in \{p_1, \dots, p_{52}\}^2$ ,

$$\frac{M_{p_j} + 1}{M_{p_i} + 1} = \varphi^{(p_j - p_i)\lambda_{\log}}.$$

*Proof.* By [Lemma 2.3](#) and the identity  $2^q/2^p = 2^{q-p}$ ,

$$\frac{2^q}{2^p} = 2^{q-p} = (\varphi^{\lambda_{\log}})^{q-p} = \varphi^{(q-p)\lambda_{\log}}.$$

Specialising to GIMPS pairs is immediate. Lean: `mersenne_ratio_golden` and `mersenne_ratio_at_known_pairs`.  $\square$

The substance of [Theorem 2.6](#) is that the entire orbit  $\{2^{p_i}\}_{i=1}^{52}$  fits into a single integer lattice  $\varphi^{\mathbb{Z}\lambda_{\log}}$  of the golden base, indexed by the gaps  $p_j - p_i \in \mathbb{Z}$ . No analogous statement holds for arbitrary  $\{r_i\} \subset \mathbb{R}_{>0}$  in place of  $\{p_i\}$ .

**Remark 2.7** (The golden lattice in binary). The ratios of [Theorem 2.6](#) have a transparent binary reading. In base 2, the Mersenne number  $M_p = 2^p - 1$  is a string of  $p$  ones (`mersenne_digits_ones`), while  $2^p = M_p + 1$  is a single bit—a 1 followed by  $p$  zeros (`two_pow_digits_single_bit`)—the increment being the carry cascade that turns  $\underbrace{1 \dots 1}_p$  into  $1 \underbrace{0 \dots 0}_p$  (`mersenne_succ_eq_two_pow`).

The golden lattice  $\{2^p\}$  is therefore exactly the set of single-bit integers, indexed by bit position, and the multiplicative step  $\varphi^{\lambda_{\log}} = 2$  is the binary left-shift (`golden_step_is_bit_shift`). The golden ratio  $2^{p_j}/2^{p_i} = \varphi^{(p_j - p_i)\lambda_{\log}}$  of [Theorem 2.6](#) is then the shift of the single bit from position  $p_i$  to position  $p_j$ .

**Corollary 2.8** (52-instance specialisation). *For each  $p \in \{p_1, \dots, p_{52}\}$ , the unique real  $\sigma$  with  $3\varphi^\sigma = 2^p$  is  $\sigma = \sigma(p_i)$ ; moreover the indices satisfy  $\sigma(p_1) < \dots < \sigma(p_{52})$ . Lean: `mersenne_phi_correspondence_52` and `sigma_strictly_increasing`.*

**Remark 2.9** (Transcendence of  $\lambda_{\log}$ ). By Gelfond–Schneider (Gelfond, 1934 [15]; Schneider, 1935 [16]), since  $\varphi$  is algebraic of degree 2 over  $\mathbb{Q}$  and  $\varphi^{\lambda_{\log}} = 2$  is algebraic of degree 1, the exponent  $\lambda_{\log}$  is transcendental over  $\mathbb{Q}$ . The identity (2.1) thus realises an exact algebraic relation between two algebraic numbers mediated by a transcendental exponent. This transcendence is formally derived in Lean 4 (`lambda_log_transcendental`), with the Gelfond–Schneider theorem itself—recently formalised by Karataakis and Wiedijk [17]—taken as the single external input.

**Corollary 2.10** (Inverse formula). *Equivalently,  $\sigma(p) = \log_\varphi(2^p/3)$  for every  $p \in \mathbb{R}$ . Lean: `sigma_mersenne_eq_logb`.*

**Remark 2.11** (The index as a golden logarithm). The inverse form exhibits the golden index  $\sigma(p)$  as a logarithm in base  $\varphi$  of an algebraic number. This is the source of its transcendence: for every integer  $p$ ,  $\varphi$  is algebraic of degree 2 and  $2^p/3$  is rational, so by the same Gelfond–Schneider argument as for  $\lambda_{\log}$  ([Remark 2.9](#)),  $\sigma(p) = \log_\varphi(2^p/3)$  is transcendental. The period  $\lambda_{\log} = \log_\varphi 2$  of [Remark 2.9](#) is the particular golden logarithm against which the whole family  $\sigma(p)$  is measured.

**Remark 2.12** (Two transcendence theorems, same architectural role). The two periods  $2\pi$  of (1.1) and  $\lambda_{\log}$  of (2.1) are both transcendental but for distinct reasons:  $\pi$  by Lindemann (1882, via the algebraicity of  $e^{i\pi} = -1$ ),  $\lambda_{\log}$  by Gelfond–Schneider (1934, via the algebraicity of  $\varphi^{\lambda_{\log}} = 2$ ). Two distinct transcendence theorems play the same structural role in two realisations of the same architecture: a transcendental period mediates between algebraic objects in each realisation.

**Corollary 2.13** (Transcendental injectivity of the golden index). *For every nonzero integer  $p$ ,  $\sigma(p)$  is transcendental over  $\mathbb{Q}$ . In particular, the 52 values  $\{\sigma(p_1), \dots, \sigma(p_{52})\} \subset \mathbb{R}$  are 52 pairwise distinct transcendental numbers, occupying 52 distinct points of the affine  $\mathbb{Z}$ -lattice  $\mathbb{Z}\lambda_{\log} - \log_\varphi 3 \subset \mathbb{R}$ .*



*Proof.* By [Corollary 2.10](#),  $\sigma(p) = \log_\varphi(2^p/3)$ . For nonzero integer  $p$ ,  $2^p/3$  is a nonzero rational, and  $\varphi$  is algebraic of degree 2 with  $\varphi \notin \{0, 1\}$ . By Gelfond–Schneider,  $\varphi^s$  is transcendental for every algebraic irrational  $s$ ; since  $\varphi^{\sigma(p)} = 2^p/3$  is rational,  $\sigma(p)$  cannot be algebraic irrational, so it is either rational or transcendental. It is not rational: if  $\sigma(p) = m/n$  with  $m \neq 0$ , then  $\varphi^m = (2^p/3)^n \in \mathbb{Q}$ , but  $\varphi^m = (L_m + F_m\sqrt{5})/2$  with  $F_m \neq 0$  for  $m \neq 0$ , so  $\varphi^m \notin \mathbb{Q}$ ; and  $m = 0$  would give  $2^p = 3$ , impossible for integer  $p$ . Hence  $\sigma(p)$  is transcendental.

Pairwise distinctness of  $\sigma(p_1), \dots, \sigma(p_{52})$  follows from the strict monotonicity of [Corollary 2.8](#) (a consequence of  $\lambda_{\log} > 0$ ).  $\square$

This is the quantitative-arithmetic content of the lattice embedding: each of the 52 golden coordinates is itself a transcendental, not just the lattice as a whole, and no algebraic choice of  $\lambda$  could produce such a configuration. The Mersenne sublattice  $\varphi^{\mathbb{Z}\lambda_{\log} - \log_\varphi 3} \subset \mathbb{R}_{>0}$  is genuinely a lattice of  $\mathbb{Z}$ -rank equal to the number of distinct Mersenne exponents, with all 52 points transcendental.

**Remark 2.14** (Numerical versus algebraic). A numerical version stating  $|3\varphi^{\sigma(p)} - 2^p| < 10^{-14}$  over the 52 known exponents is verifiable in double-precision arithmetic. [Theorem 2.2](#) is strictly stronger: it is exact, holds for all real  $p$ , and is independent of floating-point resolution.

### 3. THE FACTOR 3: THE TRIPLE COINCIDENCE

**Theorem 3.1** (Modular concentration). *For every prime  $p$ ,  $M_p \bmod 20 \in \{3, 7, 11\}$ , a subset of cardinality  $3 = M_2$  of the eight residue classes  $(\mathbb{Z}/20\mathbb{Z})^\times = \{1, 3, 7, 9, 11, 13, 17, 19\}$ . Lean: `mersenne_concentration_general`.*

*Proof.* For  $p \geq 5$  prime,  $p$  is odd and  $p \bmod 4 \in \{1, 3\}$ . Two elementary facts in  $\mathbb{Z}/20\mathbb{Z}$ :  $2^4 \equiv 16$  and  $16^k \equiv 16$  for all  $k \geq 1$ . Writing  $p = 4\lfloor p/4 \rfloor + r$  with  $r \in \{1, 3\}$ ,  $2^p = 16^{\lfloor p/4 \rfloor} \cdot 2^r \equiv 16 \cdot 2^r \pmod{20}$ , so  $2^p \bmod 20 \in \{12, 8\}$  and  $M_p \bmod 20 \in \{11, 7\}$ . The cases  $p = 2, p = 3$  give  $M_2 = 3, M_3 = 7$  directly; combining,  $M_p \bmod 20 \in \{3, 7, 11\}$ .  $\square$

**Remark 3.2** (Relation to prior literature). The concentration of [Theorem 3.1](#) is a corollary of the stronger classification of Acevedo Agudelo [18]:  $M_p \bmod 360 \in \{3, 7, 31, 127, 247, 271\}$  for every prime  $p$ , which reduces modulo 20 to the present statement. Partial precedents in the OEIS comments for sequence A000668 record weaker congruences ( $M_p \equiv 1 \pmod{6}$ ,  $M_p \equiv 7 \pmod{12}$ ,  $M_p \equiv 7 \pmod{24}$ ), none of which alone implies the mod-20 concentration. The role of [Theorem 3.1](#) in the present note is as clause (vi) of [Theorem 3.5](#), conjoined with clauses (vii) and (xi).

The restriction of all  $M_p$  to 3 of the 8 residue classes coprime with 20 is not visible from the definition of the Mersenne sequence; it is a phenomenon of the modular doubling pattern  $2^p \bmod 20$  produced by the proof above. The numerator and denominator of the ratio  $3/8 = M_2/|(\mathbb{Z}/20\mathbb{Z})^\times|$  that records this concentration quantitatively are themselves clauses (vii) and (ix) of [Theorem 3.5](#).

**Theorem 3.3** (The factor 3 is structurally  $M_2$ ). *The natural number 3 admits simultaneously the three identifications*

$$3 = M_2 = 2^2 - 1 = |\{3, 7, 11\}|.$$

*Proof.* The chain  $3 = M_2 = 2^2 - 1 = |\{3, 7, 11\}|$  comprises three identifications. First,  $M_2 = 2^2 - 1$  by definition of the Mersenne sequence, and  $2^2 - 1 = 3$  by computation. Second,  $|\{3, 7, 11\}| = 3$  by counting. Third, that  $\{3, 7, 11\}$  is exactly the set of residue classes containing  $M_p \bmod 20$  for some prime  $p$  is [Theorem 3.1](#). Lean: `three_eq_M2_eq_admissible_card` and `admissible_classes_card_eq_M2`.  $\square$

The asymmetry between 2 and 3 in [Theorem 2.2](#) is only apparent: 2 enters as the base of  $2^p$  (the substrate), while 3 enters as the multiplier. The triple identification of [Theorem 3.3](#) shows that the choice of multiplier is not free. Replacing 3 by any other positive real  $c$  in the identity produces  $c\varphi^{p\lambda_{\log} - \log_\varphi c} = 2^p$ , but no other value of  $c$  admits the simultaneous identification with  $M_2$ , with

the cardinality of the admissible residue classes of  $M_p \bmod 20$ , and with the smallest Mersenne prime occurring as the residue  $M_2 \equiv 3 \pmod{20}$  in the admissible set  $\{3, 7, 11\}$ . The value  $c = 3$  is determined by the arithmetic of Mersenne in  $(\mathbb{Z}/20\mathbb{Z})^\times$ , not by the algebra of the identity.

**Remark 3.4** (Pentagonal–cyclotomic underpinning). The golden ratio admits the pentagonal identity

$$\varphi = 2 \cos(\pi/5) = \zeta_{10} + \zeta_{10}^{-1}, \quad \zeta_{10} := e^{i\pi/5}, \quad (3.1)$$

where  $\zeta_{10}^{10} = 1$ . The first equality  $\varphi = 2 \cos(\pi/5)$  derives from the Chebyshev relation  $T_5(\cos \theta) = \cos(5\theta)$  at  $\theta = \pi/5$ , which gives  $4 \cos^2(\pi/5) - 2 \cos(\pi/5) - 1 = 0$  (after factoring out the root  $\cos(\pi/5) \neq -1$ ), and selecting the positive root yields  $\cos(\pi/5) = \varphi/2$ . The second equality  $2 \cos(\pi/5) = \zeta_{10} + \zeta_{10}^{-1}$  is the real part of Euler's identity (1.1) at  $\theta = \pi/5$ :  $e^{i\pi/5} + e^{-i\pi/5} = 2 \cos(\pi/5)$ .

These two equalities, taken jointly, register four of the twelve clauses of **Theorem 3.5**. Clause (iii)  $\varphi = 2 \cos(\pi/5)$  is the real-trigonometric embodiment of the algebraic object: the same  $\varphi$  defined by  $\varphi^2 = \varphi + 1$ , which carries no intrinsic notion of rotation or periodicity, here appears inscribed in  $\mathbb{R}$  through cosine. Clause (i)  $\varphi_{\mathbb{C}} = \zeta_{10} + \zeta_{10}^{-1}$  is the cyclotomic-complex embodiment: the same algebraic object now appears as a concrete point of Euler's structure—the sum of two conjugate vertices of the regular decagon  $\{\zeta_{10}^k\}_{k=0}^9$  in  $S^1$ . Clause (ii)  $\zeta_{10}^{10} = 1$  places  $\zeta_{10}$  not as a free point of  $S^1$  but as a generator of the cyclic group of order 10; without it  $\zeta_{10}$  would admit no Galois structure. Equivalently,  $\varphi$  inhabits the real maximal subfield  $\mathbb{Q}(\varphi) \subset \mathbb{Q}(\zeta_{20})$  of the cyclotomic field whose Galois group is

$$\text{Gal}(\mathbb{Q}(\zeta_{20})/\mathbb{Q}) \cong (\mathbb{Z}/20\mathbb{Z})^\times,$$

an abelian group of order  $\phi(20) = 8$ ; this is clause (ix), the cardinality of the Galois receptacle within which the Mersenne residues of **Theorem 3.1** are hosted. The four clauses (i), (ii), (iii), (ix) present the same algebraic object  $\varphi$  in four canonical inscriptions inside the structure of (1.1). The modular concentration of **Theorem 3.1** into 3 classes of  $(\mathbb{Z}/20\mathbb{Z})^\times$ , together with the identification  $3 = M_2$  of **Theorem 3.3**, are the arithmetic shadows of this geometric inscription. Formal identities: `phi_eq_two_cos_pi_fifth`, `phi_eq_zeta10_sum`, `zeta10_pow_ten_eq_one`.

**Theorem 3.5** (Simultaneous identities theorem for  $\varphi$ ). *A single  $\varphi$  satisfies simultaneously the following twelve identities, inclusions, and identifications in their respective canonical structures:*

- (i) Cyclotomic inscription in  $\mathbb{C}$ :  $\varphi_{\mathbb{C}} = \zeta_{10} + \zeta_{10}^{-1}$ , with  $\zeta_{10} := e^{i\pi/5}$  (**Remark 3.4**; Lean: `phi_eq_zeta10_sum`).
- (ii) Tenth-root closure:  $\zeta_{10}^{10} = 1$  (Lean: `zeta10_pow_ten_eq_one`).
- (iii) Pentagonal trigonometric identity in  $\mathbb{R}$ :  $\varphi = 2 \cos(\pi/5)$  (**Remark 3.4**; Lean: `phi_eq_two_cos_pi_fifth`).
- (iv) Logarithmic binary bridge:  $\varphi^{\lambda_{\log}} = 2$  (**Lemma 2.3**; Lean: `mersenne_bridge_via_lambda`).
- (v) Universal real-multiplicative identity: for every  $p \in \mathbb{R}$ ,  $3\varphi^{\sigma(p)} = 2^p$  (**Theorem 2.2**; Lean: `golden_tower_bridge`).
- (vi) Modular concentration: for every prime  $p$ ,  $M_p \bmod 20 \in \{3, 7, 11\}$  (**Theorem 3.1**; Lean: `mersenne_concentration_general`).
- (vii) Triple ternary identification:  $3 = M_2 = |\{3, 7, 11\}|$  (**Theorem 3.3**; Lean: `three_eq_M2_eq_admissible_card`).
- (viii) Real-complex cross identification:  $\iota(\varphi) = \varphi_{\mathbb{C}}$ .
- (ix) Galois order:  $|(\mathbb{Z}/20\mathbb{Z})^\times| = 8$  (Lean: `ZtwentyStar_card`).
- (x) Galois sub-object inclusion:  $\{3, 7, 11\} \subseteq (\mathbb{Z}/20\mathbb{Z})^\times$ .
- (xi) Quantitative ratio:

$$\frac{|\{3, 7, 11\}|}{|(\mathbb{Z}/20\mathbb{Z})^\times|} = \frac{M_2}{8} = \frac{3}{8},$$

equivalently  $|\{3, 7, 11\}| \cdot 8 = M_2 \cdot |(\mathbb{Z}/20\mathbb{Z})^\times|$ .

(xii) Rotor discretisation: *for every*  $k \in \mathbb{N}$ ,  $(\zeta_{10}^k)^{10} = 1$ .

*Formal synthesis:* `phi_euler_mersenne_connector`.

Clauses (viii), (x), (xi), (xii) have short proofs but each registers a structural knotting that the theorem requires.

**Clause (viii)** (real-complex cross identification) is the identification that permits speaking of *the same*  $\varphi$  across clause (iii) (trigonometric in  $\mathbb{R}$ ) and clause (i) (cyclotomic in  $\mathbb{C}$ ). Without it, the statement would not be about a single algebraic object but about distinct copies of  $\varphi$  in disjoint structures. The proof is the definitional behaviour of the canonical embedding  $\iota$  on the algebraic point  $\varphi$ .

**Clause (x)** ( $\{3, 7, 11\} \subseteq (\mathbb{Z}/20\mathbb{Z})^\times$ ) registers that the three admissible Mersenne residues of clause (vi) are not arbitrary integers but elements of the Galois group of clause (ix): the arithmetic locus of Mersenne primes is inscribed inside the cyclotomic Galois receptacle. The proof is that 3, 7, 11 are each coprime with 20.

**Clause (xi)** ( $|\{3, 7, 11\}|/|(\mathbb{Z}/20\mathbb{Z})^\times| = 3/8$ ) is the quantitative knotting: the first Mersenne prime  $M_2$  is the numerator of the fraction of Galois classes that host all Mersenne arithmetic. The proof is the direct numerical consequence of clauses (vii) ( $M_2 = 3$ ) and (ix) ( $|(\mathbb{Z}/20\mathbb{Z})^\times| = 8$ ); the substance is that the simultaneous holding of those two clauses under the same  $\varphi$  is what produces the ratio 3/8 as structural and not as numerical coincidence.

**Clause (xii)** ( $(\zeta_{10}^k)^{10} = 1$  for all  $k \in \mathbb{N}$ ) registers the canonical discretisation of the continuous rotation  $\theta \mapsto e^{2i\theta}$  of (1.1): evaluated at  $\theta = 2\pi k/10$  it produces a cyclic group of order 10. Clause (xii) is the first knotting in the statement between continuous and discrete: without it, the decagonal vertices  $\{\zeta_{10}^k\}_{k=0}^9$  are a set of ten complex numbers; with it, they are a finite cyclic group on which the Galois structure of (ix) acts. The proof is  $(\zeta_{10}^k)^{10} = (\zeta_{10}^{10})^k = 1^k = 1$  by clause (ii).

All four are subsumed in the formal synthesis `phi_euler_mersenne_connector`.

**Remark 3.6** (Reading of [Theorem 3.5](#)). No proper sub-collection of the twelve clauses constitutes the content of [Theorem 3.5](#); the content is the simultaneous holding of all twelve under a single  $\varphi$ . The clauses inhabit canonical structures of distinct kinds: complex-rotational and cyclotomic (i, ii, xii), real-trigonometric (iii), logarithmic and real-multiplicative (iv, v), arithmetic-modular over  $\mathbb{Z}/20\mathbb{Z}$  (vi, vii), cross-domain real-to-complex (viii), Galois-theoretic of the cyclotomic decagonal extension  $\mathbb{Q}(\zeta_{20})/\mathbb{Q}$  (ix, x, xi). The continuous rotational symmetry of Euler's identity (1.1), with period  $2\pi$  in  $S^1 \subset \mathbb{C}$ , discretises canonically into the ten vertices of the regular decagon  $\{\zeta_{10}^k\}_{k=0}^9$  (clause xii), of which  $\varphi$  is the sum of two conjugate vertices (clauses i, iii); the Galois group of the larger cyclotomic extension is  $(\mathbb{Z}/20\mathbb{Z})^\times$  of order 8 (clause ix); the Mersenne primes  $M_p$  occupy  $M_2 = 3$  of those 8 classes (clauses vi, vii, x, xi); and the same  $\varphi$  mediates, via the transcendental exponent  $\lambda_{\log}$  ([Remark 2.9](#)), the binary doubling  $\varphi^{\lambda_{\log}} = 2$  and the universal identity  $3\varphi^{\sigma(p)} = 2^p$  (clauses iv, v). The continuous rotor, the discrete arithmetic of Mersenne, the cyclotomic geometry of  $\mathbb{Q}(\zeta_{20})$ , and the real-multiplicative line  $\mathbb{R}_{>0}$  coincide simultaneously under  $\varphi$  across the twelve clauses of [Theorem 3.5](#). The quantitative ratio  $M_2/|(\mathbb{Z}/20\mathbb{Z})^\times| = 3/8$  of clause (xi) records that the first Mersenne prime is the numerator of the fraction of Galois classes that host the Mersenne arithmetic.

The arithmetic of the integers, in this reading, appears as a discrete shadow of the rotational symmetry of (1.1): the chain rotation  $\rightarrow$  polygonal discretisation  $\rightarrow$  Galois action  $\rightarrow$  residue classification  $\rightarrow$  Mersenne concentration formalises a geometric connection with the tools of Gelfond–Schneider and `AdjoinRoot`.

**3.1. The identity, characterized.** Read as an equality, the universal identity of [Theorem 2.2](#) is the evaluation of a definition:  $\sigma(p) := p\lambda_{\log} - \log_\varphi 3$  is chosen precisely so that  $3\varphi^{\sigma(p)} = 2^p$ . What is *not* definitional is the statement gathered here: that the constituents of the equation—the factor 3, the base  $\varphi$ , the slope  $\lambda_{\log}$ —are each pinned by a theorem *external* to the equation, and that

these separately fixed determinations are mutually consistent, with the golden coordinate uniquely determined.

**Theorem 3.7** (The identity as a consistency theorem). *The equation  $3\varphi^\sigma = 2^p$  in the unknown  $\sigma \in \mathbb{R}$  carries four determinations, each fixed independently of it, and they hold simultaneously:*

- (i) Existence.  $\sigma(p)$  solves it for every real  $p$  ([Theorem 2.2](#)).
- (ii) Uniqueness.  $\sigma(p)$  is its only real solution ([Theorem 2.5](#)).
- (iii) Transcendence. the slope  $\lambda_{\log}$  is transcendental over  $\mathbb{Q}$  ([Remark 2.9](#)).
- (iv) Forced factor.  $3 = M_2 = |\{3, 7, 11\}|$ , the number of residue classes the Mersenne primes occupy modulo 20 ([Theorem 3.1](#)), independently of the equation.

Lean: `golden_identity_characterization`.

**Remark 3.8** (What the theorem certifies). The central equation (i) is not proved against itself—that would be circular; what is proved is that it exists, is unique, and coincides with the external determinations (iii)–(iv). The uniqueness (ii) certifies that the golden coordinate  $\sigma(p)$  is well defined; the evidential weight lies in (iii) and (iv), where the slope and the factor are pinned by theorems that never mention the equation. In this reading the identity is neither an axiom nor a bare definition, but the single point at which independently established facts about  $\varphi$  agree. Formally,  $\sigma$  is the unique function with  $3\varphi^{\sigma(p)} = 2^p$  for all  $p$  (`golden_identity_is_consistent_overdetermined`).

#### 4. FIBONACCI SPLITTING OF THE MERSENNE PRIMES

The modular concentration of [Theorem 3.1](#) records *where* the Mersenne primes fall in  $(\mathbb{Z}/20\mathbb{Z})^\times$ : their residues lie in  $\{3, 7, 11\}$ . We now show *how* they fall—how each prime  $M_p$  splits in the field  $\mathbb{Q}(\sqrt{5})$  of the golden ratio. The answer is governed by a single classical congruence (Lucas, 1878) and, composed with the concentration, refines the three admissible classes into a split class and two inert classes. This is the arithmetic content behind the identification  $3 = M_2 = |\{3, 7, 11\}|$  of clause (vii): the multiplier 3 is the number of classes the Mersenne primes occupy, and the golden field determines the splitting type in each.

**4.1. Fibonacci prerequisites.** The Fibonacci sequence is  $F_0 = 0$ ,  $F_1 = 1$ ,  $F_{n+2} = F_{n+1} + F_n$ . In any commutative ring containing an element  $\alpha$  with  $\alpha^2 = \alpha + 1$ , induction on  $n$  gives the Binet-type identity

$$\alpha^{n+1} = F_{n+1}\alpha + F_n. \quad (4.1)$$

The Galois conjugate of  $\varphi$  over  $\mathbb{Q}$  is  $\bar{\varphi} = 1 - \varphi = -1/\varphi$ , with trace  $\varphi + \bar{\varphi} = 1$  and norm  $\varphi\bar{\varphi} = -1$ ; both roots of  $X^2 - X - 1$ . The discriminant of  $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$  is  $(\varphi - \bar{\varphi})^2 = 5$ , so 5 is the unique ramified prime. For an odd prime  $q \neq 5$ , the splitting of  $q$  in  $\mathbb{Q}(\sqrt{5})$  is governed by the Legendre symbol  $(q/5)$ :  $(q/5) = +1$  (split) iff  $q \bmod 20 \in \{1, 9, 11, 19\}$ , and  $(q/5) = -1$  (inert) iff  $q \bmod 20 \in \{3, 7, 13, 17\}$ . Among the admissible Mersenne classes  $\{3, 7, 11\}$ , exactly one—the class 11—is split. Lean: `fib_pow_aux`, `chi5_values`.

#### 4.2. The Fibonacci splitting congruence.

**Theorem 4.1** (Fibonacci splitting; Lucas 1878). *For every prime  $q \neq 2, 5$ ,*

$$F_q \equiv (q/5) \pmod{q},$$

*with value +1 in the split case and −1 in the inert case.* Lean: `fibonacci_splitting_general`.

**Lemma 4.2** (Split case). *Let  $q \neq 2, 5$  be prime with  $(q/5) = +1$ . Then  $X^2 - X - 1$  has a root  $\alpha \in \mathbb{F}_q$ , and  $F_q \equiv 1 \pmod{q}$ .* Lean: `fib_split_case`.

*Proof.* Since  $(q/5) = +1$ , there is  $\beta \in \mathbb{F}_q$  with  $\beta^2 = 5$ ; put  $\alpha := (1 + \beta)/2$  (defined as  $q \neq 2$ ). Then  $\alpha^2 = (6 + 2\beta)/4 = \alpha + 1$ . By Fermat,  $\alpha^q = \alpha$ , and by (4.1),  $\alpha^q = F_q\alpha + F_{q-1}$ ; the same holds for the conjugate root  $\bar{\alpha} = 1 - \alpha$ . Subtracting the two relations gives  $F_q(\alpha - \bar{\alpha}) \equiv \alpha - \bar{\alpha}$ ; since  $(\alpha - \bar{\alpha})^2 = 5 \not\equiv 0$  (as  $q \neq 5$ ) the factor is invertible, so  $F_q \equiv 1 \pmod{q}$ .  $\square$

Worked example ( $q = 11$ , split). Since  $11 \equiv 1 \pmod{5}$ ,  $(11/5) = +1$ ; in  $\mathbb{F}_{11}$  the roots of  $X^2 = X + 1$  are  $\alpha = 8$ ,  $\bar{\alpha} = 4$  (as  $8^2 = 64 \equiv 9 = 8 + 1$ ). And  $F_{11} = 89 = 11 \cdot 8 + 1$ , so  $F_{11} \equiv 1 \pmod{11}$ .

**Lemma 4.3** (Inert case). *Let  $q \neq 2, 5$  be prime with  $(q/5) = -1$ . Then  $X^2 - X - 1$  is irreducible over  $\mathbb{F}_q$ , the Frobenius  $x \mapsto x^q$  of  $\mathbb{F}_{q^2} = \mathbb{F}_q[X]/(X^2 - X - 1)$  exchanges the two roots  $\alpha, \bar{\alpha} = 1 - \alpha$ , and  $F_q \equiv -1 \pmod{q}$ . Lean: `fib_inert_case`.*

*Proof.* By quadratic reciprocity,  $(5/q) = (q/5) = -1$ , so 5 is a non-residue and  $X^2 - X - 1$  has no root in  $\mathbb{F}_q$ ; being quadratic it is irreducible, and the quotient  $\mathbb{F}_{q^2} = \mathbb{F}_q[X]/(X^2 - X - 1)$  is a field of order  $q^2$  with basis  $\{1, \alpha\}$ , where  $\alpha$  is the image of  $X$ . The Frobenius  $\text{Frob}_q: x \mapsto x^q$  is an  $\mathbb{F}_q$ -automorphism (additivity is the freshman's dream in characteristic  $q$ ; fixity of  $\mathbb{F}_q$  is Fermat), it permutes the roots of  $X^2 - X - 1$ , and it is not the identity (else all of  $\mathbb{F}_{q^2}$  would lie in  $\mathbb{F}_q$ ); hence  $\alpha^q = \bar{\alpha} = 1 - \alpha$ . By (4.1),  $\alpha^q = F_q\alpha + F_{q-1}$ , so  $(F_q + 1)\alpha + (F_{q-1} - 1) = 0$  in  $\mathbb{F}_{q^2}$ ; as  $\{1, \alpha\}$  is a basis, both coefficients vanish, giving  $F_q \equiv -1 \pmod{q}$ .  $\square$

**Remark 4.4** (Lucas–Lehmer companion). The inert case also yields  $F_{q-1} \equiv 1$ , whence  $F_{q+1} = F_q + F_{q-1} \equiv 0 \pmod{q}$ : the classical Fermat–Lucas condition  $q \mid F_{q+1}$  for inert primes, which underlies the index of appearance in the Lucas–Lehmer primality test.

Worked example ( $q = 7$ , inert). Since  $7 \equiv 2 \pmod{5}$  is a non-residue,  $(7/5) = -1$  and  $X^2 - X - 1$  is irreducible over  $\mathbb{F}_7$ . Here  $F_7 = 13 \equiv -1 \pmod{7}$ , and  $F_8 = 21 \equiv 0 \pmod{7}$ , consistent with Remark 4.4.

#### 4.3. Splitting type of the Mersenne primes.

**Theorem 4.5** (Residue class determines splitting type). *Let  $p$  be a prime with  $M_p$  also prime,  $M_p \neq 2, 5$ . Then  $M_p \bmod 20 \in \{3, 7, 11\}$ , and*

$$M_p \equiv 11 \Rightarrow F_{M_p} \equiv +1 \text{ (split)}, \quad M_p \bmod 20 \in \{3, 7\} \Rightarrow F_{M_p} \equiv -1 \text{ (inert)},$$

*all congruences modulo  $M_p$ . Lean: `mersenne_fibonacci_type`.*

*Proof.* Since  $M_p = 2^p - 1$  is odd and  $2^p - 1 = 5$  has no solution, the hypothesis  $M_p \neq 2, 5$  is automatic for every Mersenne prime. The concentration is Theorem 3.1 at  $p$ . By Theorem 4.1,  $F_{M_p} \equiv (M_p/5) \pmod{M_p}$ ; and  $(M_p/5)$  is read from  $M_p \bmod 5$ :  $11 \equiv 1$  gives  $+1$ , while  $3 \equiv 3$  and  $7 \equiv 2$  are non-residues, giving  $-1$ .  $\square$

| $p$ | $M_p$      | $M_p \bmod 20$ | $(M_p/5)$ | $F_{M_p} \bmod M_p$ | type  |
|-----|------------|----------------|-----------|---------------------|-------|
| 3   | 7          | 7              | -1        | -1                  | inert |
| 5   | 31         | 11             | +1        | +1                  | split |
| 7   | 127        | 7              | -1        | -1                  | inert |
| 13  | 8191       | 11             | +1        | +1                  | split |
| 17  | 131071     | 11             | +1        | +1                  | split |
| 19  | 524287     | 7              | -1        | -1                  | inert |
| 31  | 2147483647 | 7              | -1        | -1                  | inert |

TABLE 1. Fibonacci splitting type of  $M_p$  for the first seven Mersenne primes with  $p \geq 3$ .

Specialised to each of the 52 known Mersenne exponents, this classifies every known Mersenne prime by its splitting type; the distribution is Table 2.



| $M_p \bmod 20$ | Type in $\mathbb{Q}(\sqrt{5})$ | $F_{M_p} \bmod M_p$ | Count     |
|----------------|--------------------------------|---------------------|-----------|
| 3              | inert                          | -1                  | 1         |
| 7              | inert                          | -1                  | 19        |
| 11             | split                          | +1                  | 32        |
| <b>total</b>   |                                |                     | <b>52</b> |

TABLE 2. Splitting type in  $\mathbb{Q}(\sqrt{5})$  of each of the 52 known Mersenne primes, by residue class  $M_p \bmod 20$ . Verified in Lean by `native_decide` for  $p \leq 31$  and derived from `mersenne_concentration_general` otherwise.

*Explicit lists.* The single exponent in class 3 is  $p = 2$  (inert). The 19 exponents with  $M_p \equiv 7 \pmod{20}$  (inert) are 3, 7, 19, 31, 107, 127, 607, 1279, 2203, 4423, 86 243, 110 503, 216 091, 756 839, 1 257 787, 20 996 011, 24 036 583, 25 964 951, 37 156 667. The 32 exponents with  $M_p \equiv 11 \pmod{20}$  (split) are 5, 13, 17, 61, 89, 521, 2281, 3217, 4253, 9689, 9941, 11 213, 19 937, 21 701, 23 209, 44 497, 132 049, 859 433, 1 398 269, 2 976 221, 3 021 377, 6 972 593, 13 466 917, 30 402 457, 32 582 657, 42 643 801, 43 112 609, 57 885 161, 74 207 281, 77 232 917, 82 589 933, 136 279 841.

**Remark 4.6** (Classification under equivalence). Together, [Theorem 3.1](#) and [Theorem 4.5](#) realise a classification of the Mersenne primes within the pentagonal-cyclotomic geometry of  $\varphi$ : the multiplier  $3 = M_2 = |\{3, 7, 11\}|$  counts the admissible classes, and the golden field partitions them into one split class (11) and two inert classes (3, 7). The observed split/inert split—32 against 20 over the current 52—is empirical; under conjectural equidistribution one would expect a ratio nearer 1 : 2, and the deviation over a sample of size 52 is not conclusive.

**4.4. The splitting type from the exponent.** The residue  $M_p \bmod 20$  of [Theorem 4.5](#) is itself fixed by the exponent modulo 4, which reduces the whole classification to the arithmetic of  $p$  alone.

**Corollary 4.7** (Splitting type from  $p \bmod 4$ ). *Let  $M_p$  be a Mersenne prime. Its splitting type in  $\mathbb{Q}(\sqrt{5})$  is determined by  $p \bmod 4$ :*

$$p \equiv 1 \pmod{4} \Rightarrow \text{split}, \quad p \equiv 3 \pmod{4} \text{ or } p = 2 \Rightarrow \text{inert}.$$

Lean: `mersenne_mod5_from_p_mod4`.

*Proof.* The residues  $2^p \bmod 5$  have period 4, equal to 2, 4, 3, 1 for  $p \equiv 1, 2, 3, 0 \pmod{4}$ . Hence  $M_p = 2^p - 1$  modulo 5 depends only on  $p \bmod 4$ , taking the values 1, 3, 2 for  $p \equiv 1, 2, 3 \pmod{4}$ , so  $(M_p/5) = +1, -1, -1$  respectively. By [Theorem 4.5](#) this sign is the splitting type. (The case  $p \equiv 0 \pmod{4}$  gives  $5 \mid M_p$  and is excluded, since  $M_p$  is a prime  $\neq 5$ .)  $\square$

Thus, among the admissible classes, the split class 11 is exactly  $p \equiv 1 \pmod{4}$  and the inert classes 3, 7 are  $p = 2$  and  $p \equiv 3 \pmod{4}$ . The finer modulus 8 does not separate the type any further:  $p \bmod 4$  is the exact controlling residue of the exponent.

**Corollary 4.8** (Fibonacci rank characterises the inert primes). *For a Mersenne prime  $M_p \neq 5$ ,*

$$M_p \mid F_{M_p+1} \iff M_p \text{ is inert in } \mathbb{Q}(\sqrt{5}) \iff p \equiv 3 \pmod{4} \text{ or } p = 2.$$

Lean: `mersenne_rank_iff_inert`.

*Proof.* If  $M_p$  is inert then  $M_p \mid F_{M_p+1}$  by [Remark 4.4](#). If  $M_p$  is split, the roots  $\alpha, \bar{\alpha}$  of  $X^2 - X - 1$  lie in  $\mathbb{F}_{M_p}^\times$ , so  $\alpha^{M_p-1} = \bar{\alpha}^{M_p-1} = 1$  by Fermat, giving  $F_{M_p-1} = (\alpha^{M_p-1} - \bar{\alpha}^{M_p-1})/(\alpha - \bar{\alpha}) \equiv 0$ ; with  $F_{M_p} \equiv 1$  ([Lemma 4.2](#)) this yields  $F_{M_p+1} = F_{M_p} + F_{M_p-1} \equiv 1 \not\equiv 0 \pmod{M_p}$ . The second equivalence is [Corollary 4.7](#).  $\square$

The divisibility  $M_p \mid F_{M_p+1}$ —that is, the rank of apparition of  $M_p$  in the Fibonacci sequence dividing  $M_p + 1$ —therefore singles out *exactly* the inert half of the Mersenne primes, the same rank-of-apparition structure that underlies the Lucas–Lehmer test.



**Corollary 4.9** (The joint presentation from the exponent). *For a Mersenne prime  $M_p$ , the exponent residue  $p \bmod 4$  fixes the triple (residue class modulo 20, splitting type in  $\mathbb{Q}(\sqrt{5})$ , Fibonacci sign  $F_{M_p} \bmod M_p$ ) at once:*

$$\begin{aligned} p \equiv 1 \pmod{4} &\mapsto (11, \text{split}, +1), \\ p \equiv 3 \pmod{4} &\mapsto (7, \text{inert}, -1), \\ p = 2 &\mapsto (3, \text{inert}, -1). \end{aligned}$$

*Over the 52 known Mersenne primes the triple takes exactly these three values, one per admissible residue; given  $p \bmod 4$ , no further datum distinguishes the class, the type, and the sign.*

*Proof.* The class follows from the modular concentration ([Theorem 3.1](#)) together with  $M_p \bmod 5$ , which  $p \bmod 4$  fixes ([Corollary 4.7](#)); the type from the same residue; and the Fibonacci sign from quadratic reciprocity, since  $X^2 - X - 1$  has a root modulo  $M_p$  exactly when 5 is a square there, i.e. when  $M_p \equiv \pm 1 \pmod{5}$ , whence [Lemma 4.2](#) or [Lemma 4.3](#) applies.  $\square$

Lean: `mersenne_presentation_from_p_mod4`.

## 5. THE RECIPROCAL TURN $\varphi^{-\lambda_{\log}} = \frac{1}{2}$

The transcendental period  $\lambda_{\log} = \log_{\varphi} 2$  has two directions. Forward it closes the binary bridge  $\varphi^{\lambda_{\log}} = 2$  of clause (iv) ([Lemma 2.3](#)); read backward it closes the reciprocal turn  $\varphi^{-\lambda_{\log}} = \frac{1}{2}$ . This value is not a thirteenth clause but a point already inscribed in the identity, legible at three of its nodes: it is the reciprocal of clause (iv), the  $p = -1$  node of the universal identity (v) ([Theorem 2.2](#)), and the multiplicative-line image of the conjugate root of the rotor (clause (i)). We establish the three, then note where the value leads.

**Lemma 5.1** (Reciprocal face of the binary bridge).  $\varphi^{-\lambda_{\log}} = \frac{1}{2}$ .

*Proof.* By clause (iv) ([Lemma 2.3](#)),  $\varphi^{\lambda_{\log}} = 2$ ; hence  $\varphi^{-\lambda_{\log}} = 2^{-1} = \frac{1}{2}$ . In Lean this is the inverse of `mersenne_bridge_via_lambda`.  $\square$

**Corollary 5.2** (The reciprocal turn is the universal identity at  $p = -1$ ). *The reciprocal turn  $\varphi^{-\lambda_{\log}} = \frac{1}{2}$  is the value of the universal identity (v) ([Theorem 2.2](#)) at  $p = -1$ .*

*Proof.* The universal identity reads  $3 \varphi^{\sigma(p)} = 2^p$  with  $\sigma(p) = p \lambda_{\log} - \log_{\varphi} 3$ . At  $p = -1$  the binary side is  $2^{-1} = \frac{1}{2}$ , and the golden side reduces as

$$3 \varphi^{\sigma(-1)} = 3 \varphi^{-\lambda_{\log}} \varphi^{-\log_{\varphi} 3} = 3 \varphi^{-\lambda_{\log}} \cdot \frac{1}{3} = \varphi^{-\lambda_{\log}},$$

so (v) at  $p = -1$  reads  $\varphi^{-\lambda_{\log}} = \frac{1}{2}$ .  $\square$

The factor  $3 = M_2$  (clause (vii)) that divides out here is the same multiplier the identity carries at every  $p$ —the cancellation  $3 \varphi^{-\log_{\varphi} 3} = 1$  is the mechanism of (v), not a coincidence of  $p = -1$ ; what is proper to  $p = -1$  is only that the binary side reads  $\frac{1}{2}$ . The reciprocal turn therefore sits inside the central identity (v), the identity that carries the ternary factor  $3 = M_2$ .

**Remark 5.3** (Inversion image of the rotor). Clause (i) inscribes  $\varphi$  as a root and its inverse,  $\varphi = \zeta_{10} + \zeta_{10}^{-1}$ . Clause (iv) carries the same pattern—a value and its inverse—to the multiplicative line: the pair  $(\zeta_{10}, \zeta_{10}^{-1})$  on the unit circle corresponds to  $(\varphi^{\lambda_{\log}}, \varphi^{-\lambda_{\log}}) = (2, \frac{1}{2})$  on  $\mathbb{R}_{>0}$ . The rotor is closed under inversion ( $e^{i\theta} \leftrightarrow e^{-i\theta}$ ); the reciprocal turn  $\frac{1}{2}$  is, on  $\mathbb{R}_{>0}$ , the analogue of the conjugate root  $\zeta_{10}^{-1}$ , completing the inversion symmetry of the multiplicative companion so that it mirrors the rotor it accompanies.

These three inscriptions are internal: each holds by the machinery of the present identity, with no operator invoked. This same value  $\frac{1}{2}$  is where the companion operator paper anchors a self-adjoint operator in the Hilbert–Pólya programme (§ 1.2); that operator and its relation to the zeros of  $\zeta$  are developed there.

## 6. DISCUSSION

**Identity and classification under equivalence.** The organising principle of modern and contemporary mathematics is classification up to structural sameness: objects are known not in themselves but through the isomorphisms that relate them. This has a long prehistory in the rhetorical structure of Vitruvius's *De Architectura*, whose opening dichotomy *quod significatur et quod significat*—the signified and the signifier, the *res* and the *verba*—makes the work knowable through the proportional relations its *modulus* mediates: by number, proportion, and anatomy, in the pre-Cartesian mode of interpretation Foucault describes [19, 20]. The modulus is one unit from which the proportions of the whole are read—identity generating equivalence.

Modern mathematics inherits precisely this tension. Two objects may be *equal*, literally the same, or merely *isomorphic*, the same up to a structure-preserving relabelling; classification proceeds almost always by the second. The present note is unusual in sitting at the meeting of the two. Its centre is a genuine identity—a single  $\varphi$ , not a class of isomorphic objects, satisfies twelve clauses at once—yet the clauses are organised by equivalence: the residue classes of the Galois group  $(\mathbb{Z}/20\mathbb{Z})^\times$ , the split and inert types in  $\mathbb{Q}(\sqrt{5})$ . The same  $\varphi$  is read through the equivalences that classify the structures it inhabits. The arc below moves between the two registers—identity where the equation holds by construction, equivalence where the classification does the work—answering the millennial question of § 1.1 in the specific case of  $\varphi$ .

The note can be read as a single arc, from a tautology to a spectral endpoint, each step naming where it is established.

**The simultaneous setting.** The frame of the note is that a single  $\varphi$  is inscribed at once in five canonical structures—the complex rotor of Euler's identity, the real trigonometric line, the real multiplicative line, the arithmetic of Mersenne modulo 20, and the Galois group of  $\mathbb{Q}(\zeta_{20})/\mathbb{Q}$ —the twelve simultaneous clauses of Theorem 1.1. This is the golden-ratio counterpart of Euler's identity: where  $e^{i\pi} + 1 = 0$  knots analysis, algebra, and geometry at one point, here the same  $\varphi$  knots those structures through the pentagon  $\varphi = 2\cos(\pi/5)$  and its cyclotomic inscription  $\varphi = \zeta_{10} + \zeta_{10}^{-1}$ . The central identity below is  $\varphi$ 's inscription on the multiplicative line; the arc that follows traces its depth.

**The tautological identity.** Read as an equality, the central identity  $3\varphi^{\sigma(p)} = 2^p$  is tautological: the golden index  $\sigma(p) := p\lambda_{\log} - \log_{\varphi} 3$  (Definition 2.1) is defined precisely so that it holds, for every real  $p$ , and its proof (Theorem 2.2) is the evaluation of that definition. Nothing about the truth of the equation is at stake; it holds by construction. Yet the slope it carries,  $\lambda_{\log} = \log_{\varphi} 2$ , is transcendental over  $\mathbb{Q}$  (Gelfond–Schneider, Remark 2.9)—the exact counterpart of the period  $2\pi$  in Euler's identity, an algebraic relation between algebraic numbers carried by a transcendental exponent.

**Its non-tautological content.** What is not tautological is what the constituents of the equation turn out to be. The multiplier 3 is not a free constant: it equals  $M_2 = |\{3, 7, 11\}|$ , the number of residue classes the Mersenne primes occupy modulo 20 (Theorem 3.1, Theorem 3.3)—a fact about the arithmetic of  $2^p - 1$  that owes nothing to the equation. Here the identity first carries content beyond its definition.

**Its proof from outside.** That the three constituents—the factor 3, the base  $\varphi$ , the slope  $\lambda_{\log}$ —are each pinned by a theorem *external* to the identity, and are mutually consistent with the golden coordinate uniquely determined, is made precise in [Theorem 3.7](#): the identity is a consistency theorem for independently fixed determinations, certified without circularity, not a discovered equality. This is the sense in which it is proved from outside itself.

**Its expansion into the Mersenne primes.** Read forward, into the primes it produces, the identity reaches the 52 Mersenne exponents. On the multiplicative line their binary powers  $2^p = M_p + 1$  fit a single integer sublattice of the golden base, every pair standing in an exact golden ratio ([Theorem 2.6](#))—in the binary guise, the shift of the single carried bit from one position to another ([Remark 2.7](#))—with 52 distinct transcendental indices  $\sigma(p_i)$  ([Corollary 2.13](#)).

In the field  $\mathbb{Q}(\sqrt{5})$  the reach is finer, and it resolves each Mersenne prime into presentations of one invariant. The exponent residue  $p \bmod 4$  fixes at once its class in  $(\mathbb{Z}/20\mathbb{Z})^\times$ , its splitting type ([Corollary 4.7](#)), and its Fibonacci sign  $F_{M_p} \equiv \pm 1$  ([Theorem 4.1](#)):

$$p \equiv 1 \mapsto (11, \text{split}, +1), \quad p \equiv 3 \mapsto (7, \text{inert}, -1), \quad p \equiv 2 \mapsto (3, \text{inert}, -1),$$

the triple recording the class modulo 20, the type in  $\mathbb{Q}(\sqrt{5})$ , and the sign of  $F_{M_p}$ ; and the inert primes are exactly those with  $M_p \mid F_{M_p+1}$  ([Corollary 4.8](#)). Over the 52 known primes the joint presentation takes exactly three values, one per admissible residue ([Corollary 4.9](#)); given  $p \bmod 4$ , no further information separates the class, the type, and the sign. The development is [§ 4](#).

**The spectral meeting point.** Read backward, the same period  $\lambda_{\log}$  carries  $\varphi$  down to one half,  $\varphi^{-\lambda_{\log}} = \frac{1}{2}$  ([§ 5](#)). This is the value at which the companion operator paper anchors its Hilbert–Pólya operator—the point at which the golden arithmetic developed here meets the spectral programme.

Throughout, the note works on the axis of representation and equivalence rather than literal equality. What a comparison of expressions registers as distinct— $M_p$  against  $2^p$ , an index defined against one proved—the relating equivalences return to a single object in several guises. This is quantitative, not figurative: given the exponent residue  $p \bmod 4$ , the residue class, the splitting type, and the Fibonacci sign carry no information beyond one another, so that over the 52 known Mersenne primes their joint presentation takes exactly three values, one per admissible residue. The content lies not in any single presentation but in the exact maps between them; where a literal reading sees distinct objects that fail to coincide, the equivalence measures no difference at all.

## 7. FORMAL VERIFICATION IN LEAN 4

The entire argument is formalised in a single self-contained file, `golden_mersenne.lean` (1599 lines, importing only Mathlib, and including the Fibonacci splitting of [§ 4](#) and its consequences), built against Mathlib4 (Lean toolchain v4.28.0, commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365). Gelfond–Schneider is not present at this revision and enters as the explicit axiom recorded below. [Tables 3](#) and [4](#) list every Lean identifier with its verification status, grouped by role.

The external assumption set reduces to two items: (i) the Gelfond–Schneider theorem, declared as `transcendental_cpow_of_isAlgebraic_of_irrational`, from which the transcendence of  $\lambda_{\log}$  over  $\mathbb{Q}$  is *derived* (`lambda_log_transcendental`, [Remark 2.9](#)); and (ii) 44 `axiom` declarations  $M_{61}, \dots, M_{82589933}$  prime (Lucas–Lehmer, GIMPS). The golden relation  $\varphi^2 = \varphi + 1$  and the irrationality of  $\lambda_{\log}$  are proved, not assumed. Gelfond–Schneider was recently formalised in Lean 4 [\[17\]](#) and is being integrated into Mathlib; upon its landing the first axiom is discharged. No `sorry` appears in any proof.

## 8. CLOSING REMARK

We have proved and formally verified a simultaneous identities theorem for the golden ratio: a single  $\varphi$  satisfies simultaneously twelve identities, inclusions, and identifications across five canonical structures—the complex rotor of Euler’s identity, the real trigonometric line, the real-multiplicative line, the arithmetic of Mersenne modulo 20, and the Galois group of the cyclotomic extension  $\mathbb{Q}(\zeta_{20})/\mathbb{Q}$ . The central clause is the universal identity  $3\varphi^{\sigma(p)} = 2^p$ , a mediation between the binary base of Mersenne arithmetic, the ternary base  $3 = M_2$ , and the golden base of pentagonal–cyclotomic geometry through the bridge constant  $\lambda_{\log} = \log_{\varphi} 2$ . The 52 currently known Mersenne prime exponents fit into this structure as a distinguished integer sublattice of  $\varphi^{\mathbb{Z}\lambda_{\log} - \log_{\varphi} 3}$ , and every pair of Mersenne numbers admits an exact golden ratio.

## USES OF GENERATIVE ARTIFICIAL INTELLIGENCE

Generative AI tools were used by the authors as assistants in the following parts of the preparation of this work: drafting and editing the English prose; bibliographic search and cross-checking of references; assistance in writing and debugging the Lean 4 formalisation; assistance in writing the numerical verification scripts (mpmath/sympy); and organisation of the L<sup>A</sup>T<sub>E</sub>X source. All mathematical results, definitions, and proofs are the authors’ own, and the entire manuscript has been extensively reviewed and corrected by the human authors. The correctness of every formal claim is guaranteed by the Lean 4 kernel, which verifies each theorem against its axioms independently of how the proof was produced; the corresponding source compiles with 0 `sorry`. The authors take full responsibility for the accuracy, originality, and integrity of the manuscript.

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## CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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| Clause                            | Statement                                                 | Lean identifier                | Status  |
|-----------------------------------|-----------------------------------------------------------|--------------------------------|---------|
| <i>Clauses of Theorem 1.1</i>     |                                                           |                                |         |
| (i)                               | $\varphi = \zeta_{10} + \zeta_{10}^{-1}$                  | phi_eq_zeta10_sum              | 0 sorry |
| (ii)                              | $\zeta_{10}^{10} = 1$                                     | zeta10_pow_ten_eq_one          | 0 sorry |
| (iii)                             | $\varphi = 2 \cos(\pi/5)$                                 | phi_eq_two_cos_pi_fifth        | 0 sorry |
| (iv)                              | $\varphi^{\lambda_{\log}} = 2$                            | mersenne_bridge_via_lambda     | 0 sorry |
| (v)                               | $3\varphi^{\sigma(p)} = 2^p \ (\forall p \in \mathbb{R})$ | golden_tower_bridge            | 0 sorry |
| (vi)                              | $M_p \bmod 20 \in \{3, 7, 11\}$<br>( $\forall p$ prime)   | mersenne_concentration_general | 0 sorry |
| (vii)                             | $3 = M_2 =  \{3, 7, 11\} $                                | three_eq_M2_eq_admissible_card | 0 sorry |
| (ix)                              | $ (\mathbb{Z}/20\mathbb{Z})^\times  = 8$                  | ZtwentyStar_card               | 0 sorry |
| (viii),<br>(x),<br>(xi),<br>(xii) | structural knottings articulated after Theorem 3.5        | phi_euler_mersenne_connector   | 0 sorry |
| <i>Auxiliary results (§ 2)</i>    |                                                           |                                |         |
| —                                 | $\varphi^2 = \varphi + 1$ (defining relation)             | phi_sq                         | 0 sorry |
| —                                 | $\varphi^{\log_\varphi 3} = 3$ (Lemma 2.4)                | phi_pow_log3                   | 0 sorry |
| —                                 | Uniqueness of $\sigma(p)$ (Theorem 2.5)                   | sigma_mersenne_unique          | 0 sorry |
| —                                 | $2^q/2^p = \varphi^{(q-p)\lambda_{\log}}$ (Theorem 2.6)   | mersenne_ratio_golden          | 0 sorry |
| —                                 | 52-instance identity (Corollary 2.8)                      | mersenne_phi_correspondence_52 | 0 sorry |
| —                                 | Strict order of indices (Corollary 2.8)                   | sigma_strictly_increasing      | 0 sorry |
| —                                 | Irrationality of $\lambda_{\log}$                         | lambda_log_irrational          | 0 sorry |
| —                                 | Transcendence of $\lambda_{\log}$ (Remark 2.9)            | lambda_log_transcendental      | 0 sorry |
| <i>Euler rotor (Eq. (1.1))</i>    |                                                           |                                |         |
| —                                 | $e^{i\pi} = -1$ (rotor)                                   | euler_rotor                    | 0 sorry |

TABLE 3. Lean 4 identifiers and verification status (part I): clauses of the simultaneous identities theorem, auxiliary results, and the Euler rotor.



| Clause                                   | Statement                                           | Lean identifier                                               | Status               |
|------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------|----------------------|
| <i>Fibonacci splitting (§ 4)</i>         |                                                     |                                                               |                      |
| —                                        | Binet identity in $\mathbb{Z}[\varphi]$             | <code>fib_pow_aux</code>                                      | 0 sorry              |
| —                                        | Base cases $p \in \{2, 3, 5\}$                      | <code>mersenne_fibonacci_tabulated</code>                     | native<br>_decide    |
| —                                        | Split/inert partition mod 5                         | <code>chi5_values</code>                                      | 0 sorry              |
| —                                        | Lucas congruence $F_q \equiv (q/5)$ (Theorem 4.1)   | <code>fibonacci_splitting_general</code>                      | 0 sorry              |
| —                                        | Split case (Lemma 4.2)                              | <code>fib_split_case</code>                                   | 0 sorry              |
| —                                        | Inert case (Lemma 4.3)                              | <code>fib_inert_case</code>                                   | 0 sorry              |
| —                                        | Mersenne splitting type (Theorem 4.5)               | <code>mersenne_fibonacci_type</code>                          | 0 sorry              |
| —                                        | Type from $p \bmod 4$ (Corollary 4.7)               | <code>mersenne_mod5_from_p_mod4</code>                        | 0 sorry              |
| —                                        | Rank characterises inert (Corollary 4.8)            | <code>mersenne_rank_iff_inert</code>                          | 0 sorry              |
| —                                        | Joint presentation from $p \bmod 4$ (Corollary 4.9) | <code>mersenne_presentation_from_p_mod4</code>                | 0                    |
| <i>Identity characterization (§ 3.1)</i> |                                                     |                                                               |                      |
| —                                        | Consistency & uniqueness (Theorem 3.7)              | <code>golden_identity_characterization</code>                 | 0                    |
| —                                        | Overdetermined consistency (Remark 3.8)             | <code>golden_identity_is_consistent_overdetermined</code>     | 0                    |
| <i>External inputs (axioms)</i>          |                                                     |                                                               |                      |
| —                                        | Primality of $M_p$ , $p \leq 31$                    | <code>mersenne_prime_{2..31}</code><br>thms)                  | (8 native<br>_decide |
| —                                        | Primality of $M_p$ , $p \geq 61$                    | 44 GIMPS axioms                                               | external             |
| —                                        | Gelfond–Schneider (Hilbert’s 7th)                   | <code>transcendental_cpow_of_isAlgebraic_of_irrational</code> | external             |

TABLE 4. Lean 4 identifiers and verification status (part II): the Fibonacci splitting, the identity characterization, and the external inputs.